

TM Forum Component

Product Configurator

TMFC027

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Notice

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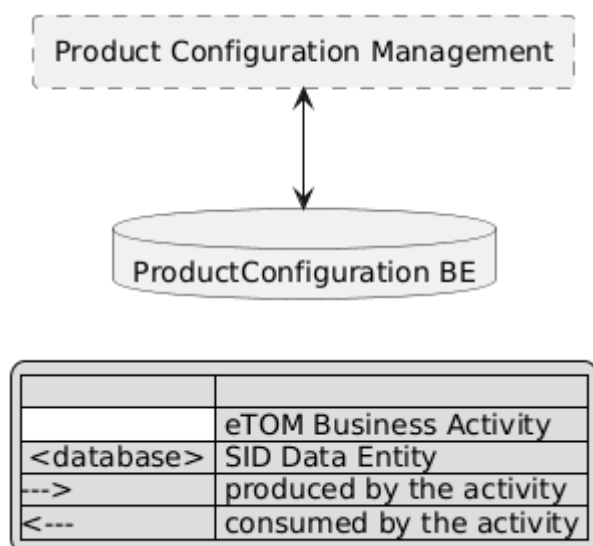
TMFC027 – Product Configurator

1. Overview

Component Name	ID	Description	ODA Function Block
Product Configurator	TMFC027	The Product Configurator component empowers sales by offering the right product combinations, controlling discounting, and automating approvals through support of configuration, pricing and quote generation activities.	Core Commerce

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.2.5	L2	Product Configuration Management	Configuration Management configures or creates a new version of a configuration for an entity, such as a product, as defined by a configuration specification. This process also modifies a configuration and values for configuration parameters, and removes a configuration.
1.2.5.2	L3	Manage Product Configuration	Manage Product Configuration business activity is in charge of creating, maintaining, controlling, changing and reporting Product Configuration according to Product Configuration Plans. / Manage Product Configuration will establish and maintaining consistency of a product's performance, functional, and physical attributes within the limits defined by product requirements, product design, and operational information throughout the Products Lifecycle.

2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition

Product Configuration	A Product Configuration (also referred to as a Product Profile) defines how a Product operates or functions. / A Product Configuration may contain one or more parts, and each part may contain zero or more fields. Each field may have attributes that are statically or dynamically defined. Some of these fields have fixed values, while others provide values from which a choice or choices can be made (e.g. using the EntitySpec/Entity and/or CharacteristicSpec/CharacteristicValue patterns) 1	ProductConfiguration	A representation of how a Product operates or functions in terms of characteristics and related Resource(s), Product(s), Service(s).
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2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
55	Price & Discount Calculation	Rating and Follow up	Tariff Calculation and Rating	Price and Discount Calculation applies pricing and discounting rules and algorithms in the context of the assembled information concerning Products (i.e. instances of Product).
182	Inter Product Dependency Identification	Product Configuration & Activation	Offer and Product Configuration	Inter-Product Dependency Identification identifies product dependencies, binds new order to purchased product or point to the dependent product required

205	Customer Order Eligibility Validation	Customer Order Management	Customer Order Eligibility Validation	Customer Order Eligibility Validation function validates that the Offer & products specified on the Customer Order, are eligible from a commercial and functional point of view. / It includes: / • Commercial Eligibility with commercial compatibility with the already customer installed Offers / • Functional Eligibility with the customer's already installed Products (corresponding to ProductSpecification).
207	Offer and Product Configuration	Product Configuration & Activation	Offer and Product Configuration	The Offer and Product Configuration function enables the configuration of the commercial offer chosen by the customer. / The configuration recovers the choice of an option, the choice of the characteristics values for the Product Specification including installation preferences... / It can be based on product configurator using a rule engine. / It can be used at the same time by Selling, Order Establishment or Develop Sales Proposal Business activities.

262	Product Availability Checking 1	Customer Order Management	Customer Order Eligibility Validation	Product Availability Checking function provide an internet technology driven interface for the customer to undertake a product availability check. / E.g., that the product offering is active for sales, the equipment(s) specified in the customer order are on stock.
274	Quote Price Support Access	Fulfillment Integration Management	Customer Fulfillment Access Management	Quote Price Support Access provides self empowered fulfillment function to provide an internet technology driven interface for the customer to get a Quotation price.
278	Customer SLA Preferences Capturing	Product Configuration & Activation	Offer and Product Configuration	Customer SLA Preferences Capturing captures the customer's SLA preferences e.g., as part of the fulfillment.

300	Discount Calculation	Rating and Follow up	Tariff Calculation and Rating	Discounts Calculation determines charge discounts based on pricing plan; including discounts on recurring, one time, and usage charges. / Discounts may be applied at different levels such as cross product, cross location, or cross customer (all customers that are part of a given group plan – some affiliation). The discounts can be apportioned across multiple events.
320	Customer Product Proposal Creation	Product Configuration & Activation	Offer and Product Configuration	Customer Product Proposal Creation proposes according to what the customer presently has as part of what can be further provided to the customer including bundling, product proposals, etc.
379	Product Customization Offering Management	Sales Management	Opportunity Management	Product Customization Offering Management provides the necessary functionality to manage the customer personalized proposals, taking into account the customer location, needs, current products, as well as the service provider's products, sales emphasis and targets, etc.

727	Product Offer to Customer Verification 2	Customer Order Management	Customer Order Eligibility Validation	Product Offer to Customer Verification enables and verifies the configuration of the commercial offer chosen by the customer. / The configuration consist of technical, functional, and commercial prerequisites and preferences.
928	Solution Design Creation	Sales Management	Opportunity Management	Solution Design Creation function combine/configure the emerged solution based on the existing solution, planned changes and newly designed features (e.g., for site connectivity services) / Selecting relevant Products and Services from Catalog – Browse and select entries from a catalog that might be relevant to the set of captured requirements into the design.
930	Automatic Solution Validation	Sales Management	Opportunity Management	Obtains configuration constraints from the catalog and validates the correctness of the design against them.

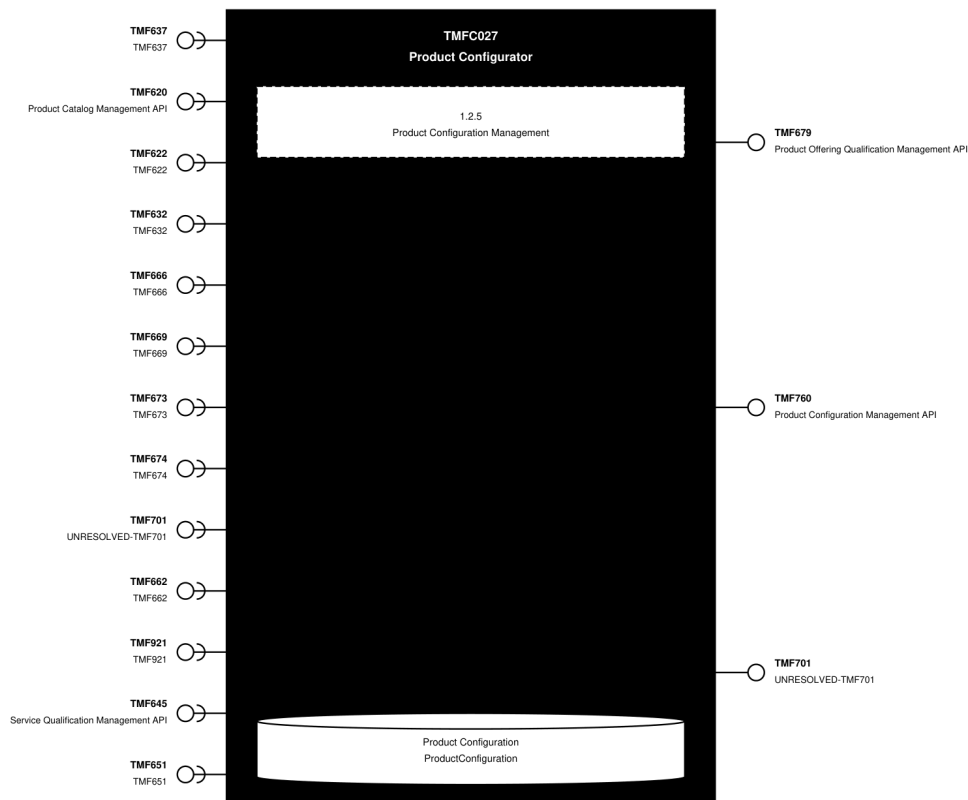
931	Solution Pricing	Sales Management	Opportunity Management	The Solution Pricing function is concerned with assuring that the designs are priced consistent with pricing used for billing. The common product catalog provides an initial price base for the components that are in the solution vs. the existing configuration at the customer location.
932	Calculation Rules Retrieval	Sales Management	Opportunity Management	Calculation Rules Retrieval function gives support for tariffication rules including discounting rules. / It supports that discounting rules and guidelines are provided as to standard levels of discounts/promotions that can be provided to the customer. Special discount arrangements can be obtained by following an escalation process. There is workflow functionality to help manage discount escalation.

933	Price/Cost Optimization	Sales Management	Opportunity Management	Price Optimization enables sales to effectively evaluate the customer, generate recommendations for price decreases and increases, and set negotiation guidelines based on our cost. This includes the application of non-standard pricing.
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3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

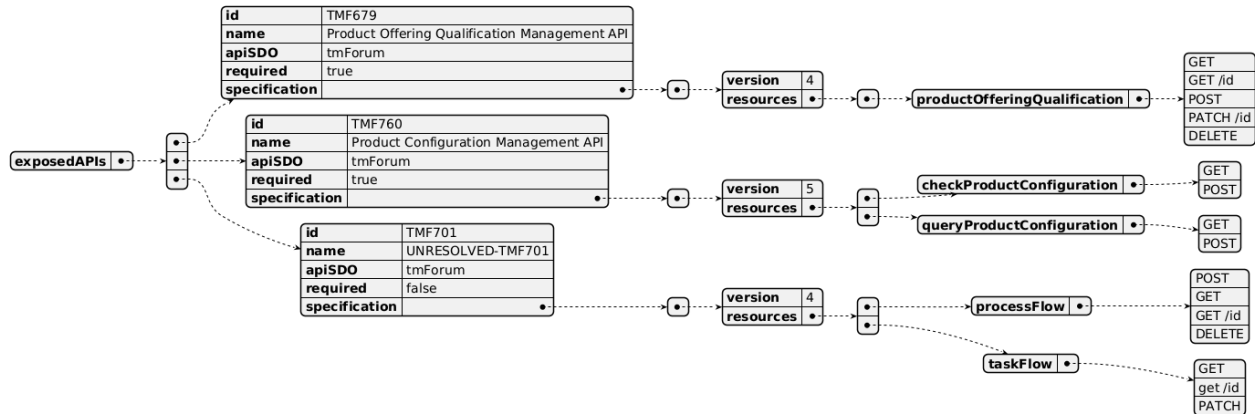


(SVG source: [TMFC027_API_Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the *component-specification-documentation skill's scripts/render_api_context_svg.py*.)

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF679	Product Offering Qualification Management API	Mandatory	4	productOfferingQualification	GET /id, POST, PATCH /id, DELETE
TMF760	Product Configuration Management API	Mandatory	5	checkProductConfiguration	GET, POST
TMF760	Product Configuration Management API	Mandatory	5	queryProductConfiguration	GET, POST

TMF701	UNRESOLVED-TMF701	Optional	4	processFlow	POST, GET, GET /id, DELETE
TMF701	UNRESOLVED-TMF701	Optional	4	taskFlow	GET, get /id, PATCH

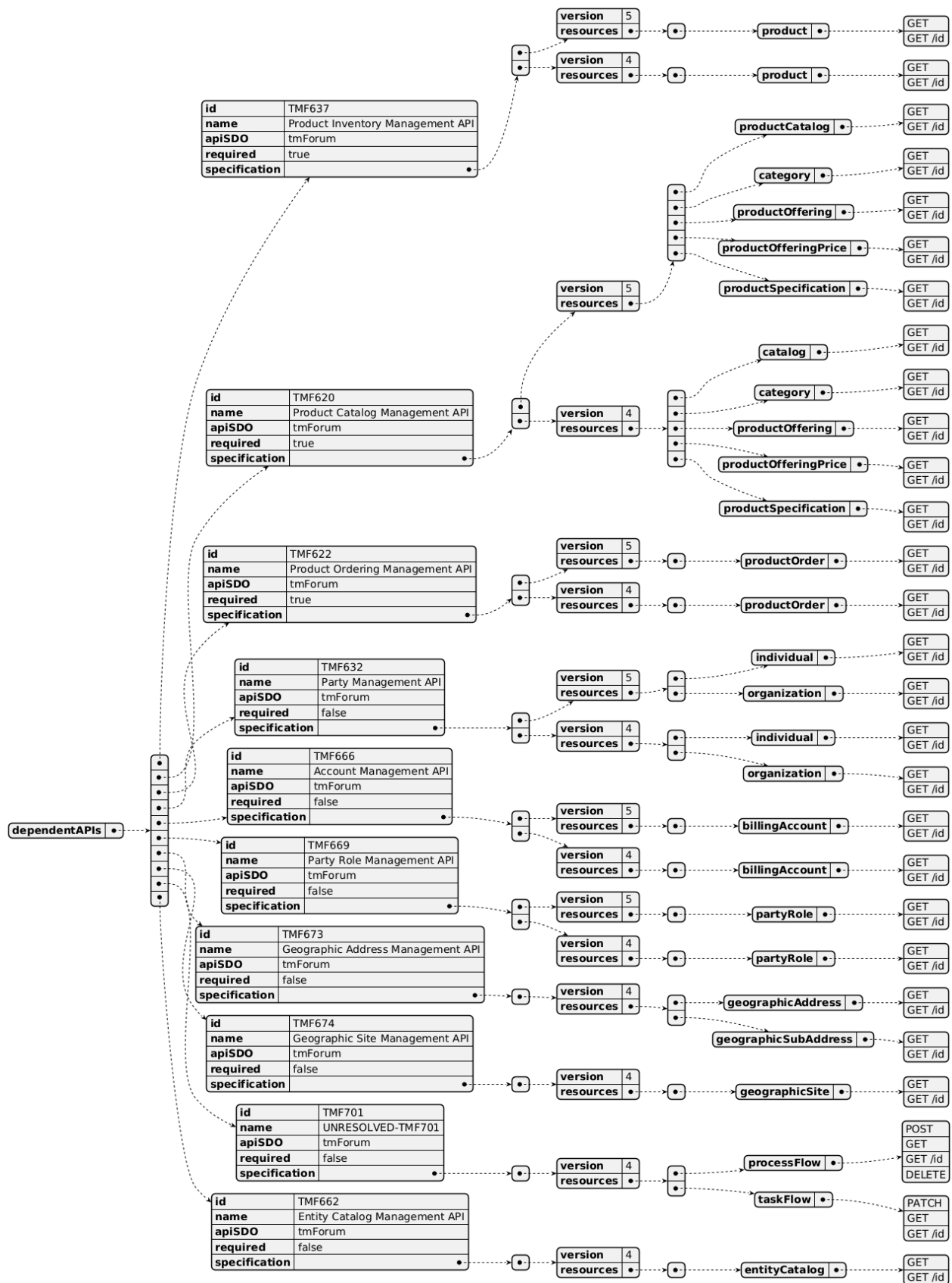


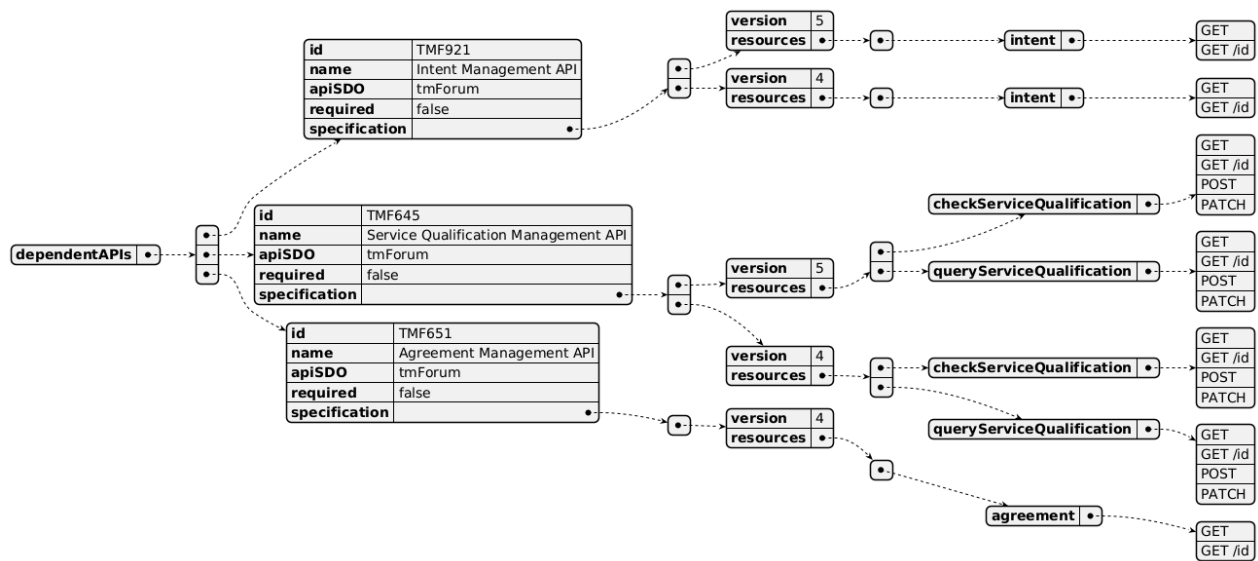
3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF637	TMF637	5	Mandatory	product	GET, GET /id
TMF637	TMF637	4	Mandatory	product	GET, GET /id
TMF620	Product Catalog Management API	5	Mandatory	productCatalog	GET, GET /id
TMF620	Product Catalog Management API	5	Mandatory	category	GET, GET /id
TMF620	Product Catalog Management API	5	Mandatory	productOffering	GET, GET /id
TMF620	Product Catalog Management API	5	Mandatory	productOfferingPrice	GET, GET /id
TMF620	Product Catalog Management API	5	Mandatory	productSpecification	GET, GET /id
TMF620	Product Catalog Management API	4	Mandatory	catalog	GET, GET /id

TMF620	Product Catalog Management API	4	Mandatory	category	GET, GET /id
TMF620	Product Catalog Management API	4	Mandatory	productOffering	GET, GET /id
TMF620	Product Catalog Management API	4	Mandatory	productOfferingPrice	GET, GET /id
TMF620	Product Catalog Management API	4	Mandatory	productSpecification	GET, GET /id
TMF622	TMF622	5	Mandatory	productOrder	GET, GET /id
TMF622	TMF622	4	Mandatory	productOrder	GET, GET /id
TMF632	TMF632	5	Optional	individual	GET, GET /id
TMF632	TMF632	5	Optional	organization	GET, GET /id
TMF632	TMF632	4	Optional	individual	GET, GET /id
TMF632	TMF632	4	Optional	organization	GET, GET /id
TMF666	TMF666	5	Optional	billingAccount	GET, GET /id
TMF666	TMF666	4	Optional	billingAccount	GET, GET /id
TMF669	TMF669	5	Optional	partyRole	GET, GET /id
TMF669	TMF669	4	Optional	partyRole	GET, GET /id
TMF673	TMF673	4	Optional	geographicAddress	GET, GET /id
TMF673	TMF673	4	Optional	geographicSubAddress	GET, GET /id
TMF674	TMF674	4	Optional	geographicSite	GET, GET /id
TMF701	UNRESOLVED-TMF701	4	Optional	processFlow	POST, GET, GET /id, DELETE
TMF701	UNRESOLVED-TMF701	4	Optional	taskFlow	PATCH, GET, GET /id
TMF662	TMF662	4	Optional	entityCatalog	GET, GET /id

TMF921	TMF921	5	Optional	intent	GET, GET /id
TMF921	TMF921	4	Optional	intent	GET, GET /id
TMF645	Service Qualification Management API	5	Optional	checkServiceQualification	GET, GET /id, POST, PATCH
TMF645	Service Qualification Management API	5	Optional	queryServiceQualification	GET, GET /id, POST, PATCH
TMF645	Service Qualification Management API	4	Optional	checkServiceQualification	GET, GET /id, POST, PATCH
TMF645	Service Qualification Management API	4	Optional	queryServiceQualification	GET, GET /id, POST, PATCH
TMF651	TMF651	4	Optional	agreement	GET, GET /id



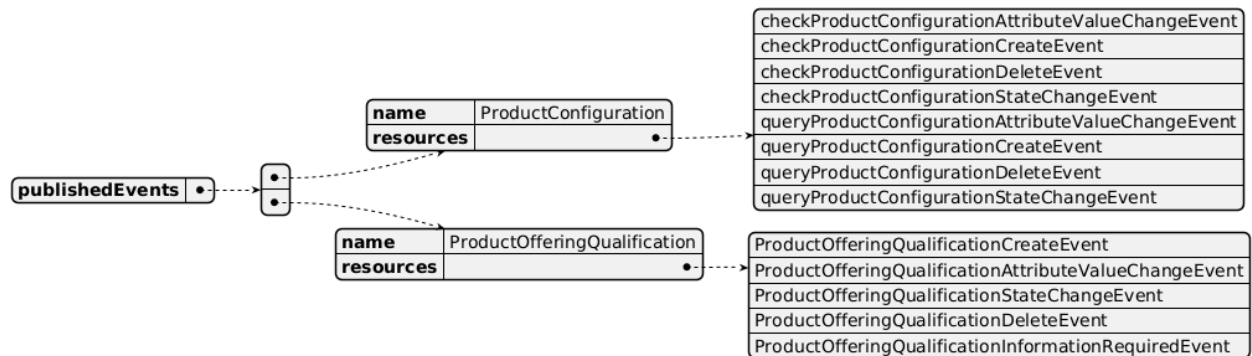


3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

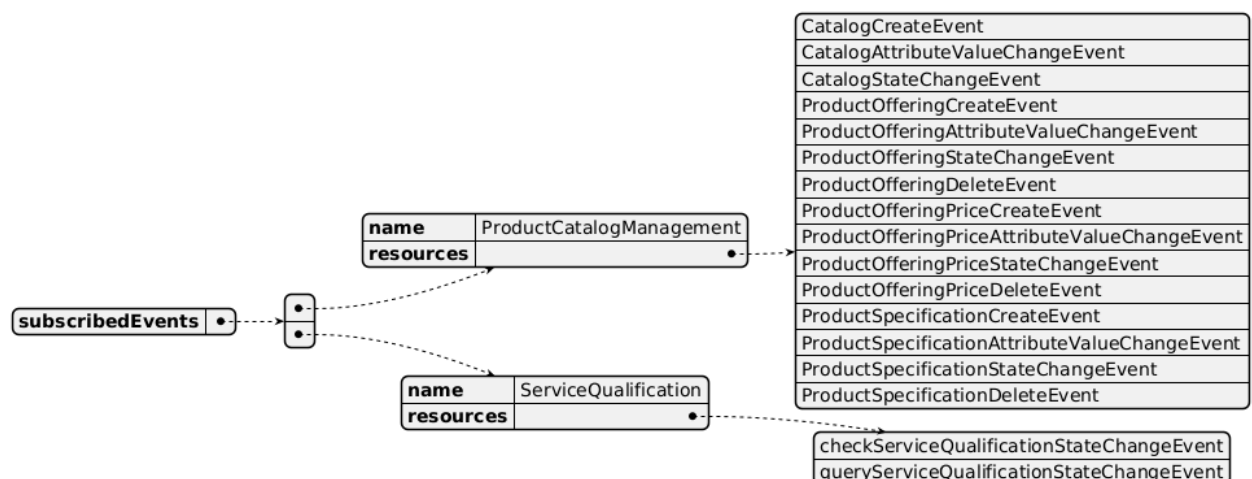
Published Events

API ID	API Name	Event Resources
TMF760	Product Configuration Management API	checkProductConfigurationAttributeValueChangeEvent checkProductConfigurationCreateEvent checkProductConfigurationDeleteEvent checkProductConfigurationStateChangeEvent queryProductConfigurationAttributeValueChangeEvent queryProductConfigurationCreateEvent queryProductConfigurationDeleteEvent queryProductConfigurationStateChangeEvent
TMF679	Product Offering Qualification Management API	ProductOfferingQualificationCreateEvent ProductOfferingQualificationAttributeValueChangeEvent ProductOfferingQualificationStateChangeEvent ProductOfferingQualificationDeleteEvent ProductOfferingQualificationInformationRequiredEvent



Subscribed Events

API ID	API Name	Event Resources
TMF620	Product Catalog Management API	CatalogCreateEvent CatalogAttributeValueChangeEvent CatalogStateChangeEvent ProductOfferingCreateEvent ProductOfferingAttributeValueChangeEvent ProductOfferingStateChangeEvent ProductOfferingDeleteEvent ProductOfferingPriceCreateEvent ProductOfferingPriceAttributeValueChangeEvent ProductOfferingPriceStateChangeEvent ProductOfferingPriceDeleteEvent ProductSpecificationCreateEvent ProductSpecificationAttributeValueChangeEvent ProductSpecificationStateChangeEvent ProductSpecificationDeleteEvent
TMF645	Service Qualification Management API	checkServiceQualificationStateChangeEvent queryServiceQualificationStateChangeEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v25.0
SID	v25.0
Functional Framework	v25.0

5.2. Jira References

No open Jira issues currently listed for this component.

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.
2. TAC-246 - Some Missing Dependent APIs (BACKLOG)

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0	29-Mar-2022	Gaetano Biancardi, Anastasios Sarantis, Sylvie Demarest	Final edits prior to publication

1.0.1	27-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	15-Aug-2023	Amaia White	Final edits prior to publication
2.0.0	10-Nov-2023	Sylvie Demarest	Added TMF760 Product Configuration as exposed API; checked dependent APIs; improved the component description; updated the mapping to Frameworks 23.0
2.1.0	24-Jun-2024	Gaetano Biancardi	Updated to latest component template; exposed API removed: TMF688 Event Mgmt, TMF672 User Roles and Permissions; dependent API removed: TMF688 Event Mgmt
2.1.1	18-Nov-2024	Gaetano Biancardi	API version, only major version to be specified
2.1.1	27-Dec-2024	Rosie Wilson	Final edits prior to publication
2.1.2	15-Sep-2025	Rosie Wilson	Updates to description per agreed YAML updated on 03-Jun-2025; final administrative updates prior to publication
2.2.0	10-Mar-2026	Gaetano Biancardi	Updated dependent APIs, added: TMF651 Agreement Mgmt, TMF645 Service Qualification Mgmt
2.2.0	24-Mar-2026	Elisabeth Andersson	Updated description to clarify that the component is not responsible for quote generation itself but provides input; removed annotation #2 under Functional Framework as this component now takes responsibility for technical qualification checks
2.2.0	28-Mar-2026	Amaia White	Final edits before publication
2.2.1	05-Jun-2026	Hugo Vaughan (TM Forum)	Gen 5 APIs added
2.2.1	26-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	29-Mar-2022	Goutham Babu	Initial release
Pre-production	15-Aug-2023	Amaia White	New release 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	20-Dec-2023	Amaia White	New release 2.0.0
Production	09-Feb-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	12-Jul-2024	Amaia White	New release 2.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	27-Dec-2024	Rosie Wilson	New release 2.1.1
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	15-Sep-2025	Rosie Wilson	New release 2.1.2
Production	14-Nov-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	28-Mar-2026	Amaia White	New release 2.2.0
Pre-production	26-Jun-2026	Sannah Sibaya	Final edits prior to publishing

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team.

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